Project "Cogni" - 2026

Department: Software Engineering

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Document Control

Item	Details
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Distribution	Software Engineering Department

Part 1 - Project Scope

1.1 Introduction

ProLeaf provides intuitive educational software for teachers and parents. Phase 1 (the secure backend foundation) is complete. Phase 2 is the frontend, AI integration, and live product that our first waitlist customers (parents and teachers) will use.

1.2 Objectives

Cogni Model - (Teachers)

- Reduce Administrative Burden: The AI Agent must generate a usable draft lesson plan and a live class checklist, aiming to save the teacher at least 30 minutes per class period on prep and admin work.

- Enable True Multitasking: The "Quadra Technology" intuitive bar and split-screen workspace must allow teachers to monitor student data, manage a lesson timeline, and access resources without switching tabs or leaving the classroom view.

- Restore Calm to the Classroom: The interface must feel intuitive and non-intrusive, reducing the cognitive load on teachers during live instruction.

ProLeaf Model - (Parents)

- Provide Transparency & Peace of Mind: Parents must be able to log in, bind to their child securely, and view their progress and comparison data in a clear, beautiful format within seconds.

- Build Trust Through Security: Parents must feel confident that their child's data is protected by real authorization boundaries (404s, not 403s) and audit trails.

- Simplify Consent: Parents must consent be able to manage their consent with a clear understanding of the policy version they are agreeing to.

Technical and Engineering Objectives

- ProLeaf Intelligence Integration, ensuring it is able generate lesson plans and checklists. The basic components of ProLeaf Intelligence is Our Agent + DeepSeek V4 Flash + Cloud Database (MLPS 2.0 Level 3).

- Connect the Frontend to the Backend: Achieve seamless end-to-end communication between the Flutter client and the Go API for all core flows (auth, teacher entry, parent view).

- Maintain the Security Standard: Ensure all Phase 1 authorization, consent, and audit tests continue to pass as new features are added. No regression in data privacy.

- Deploy to Production: Move the service from local Docker to a secure cloud environment with real direct notification delivery, production secrets management, and a public URL.

- Achieve Real-Time Responsiveness: Implement a mechanism (WebSocket or polling) so the live classroom checklist updates in the Quadra bar without requiring a page refresh.

Design and Experience Objectives

- Perfect the "Quadra Technology" Interaction: The adaptive toolbar must expand, collapse, and transition between states smoothly, surpassing the premium feel of Apple's "Dynamic Island".

- Implement the "Motion Fluid": The floating white square must allow teachers to scrub between chart types (bar, line, pie) with tactile, immediate feedback.

- Deliver a Consistent Visual Identity: The soft pink gradients, glassmorphism, and black "pill" aesthetic must be faithfully reproduced in the Flutter app.

- Ensure Accessibility: Fix the potential contrast issues during the development process identified in the design phase so the app is readable in all lighting conditions.

Constraints and Non-Objectives (What We Are Not Doing)

- Not supporting more than two languages (English and Mandarin)

- Not building predictive analytics ("This Student Will Fail") - only historical data.

- Not building offline mode - internet connection required.

1.3 - Assigned Members

Members	Role
罗江华 - Frontend Developer & Project Lead	- Build the Flutter Client from Scratch - Implement "Quadra Technology" - Build "Motion Fluid" - Develop the Split-Screen Workspace - Integrate the AI Agent UI - Connect to the Backend API - Ensure Design Fidelity - Fix Accessibility Issues - Optimize for Performance - Collaborate with Backend - Write Tests
Rasal Amarisuriya - Backend Developer	- Harden the Core (Penetration Testing, Rate Limiting, Token Security Review, Database Query Optimization, Backup and Recovery, Monitoring and Reliability). - Database schema and all migrations - API contract and documentation - Admin API - Core Compliance (data localization, encryption, retention) - Define Intelligence Interface Contract - Compliance Backend Support - Fallback Logic - Cost-tracking infrastructure - CSV Import Pipeline - AI Integration Support

苏越 - Junior Backend Developer	- Learn the Codebase - AI Integration (ProLeaf Intelligence + DeepSeek V4 Flash) - AI endpoint integration support - ICP Filing and Algorithm Filing - Stats / Data Pipeline - Compliance Infrastructure - CSV Import Support

1.4 Official Deadline

Phase	Dates	Duration	Goal
Phase 1 (Core-Build)	2026.09.28 - 2027.01.11	3.5 months, 105 days	- Feature-complete. All core features working. Ready for QA.
Phase 2 (Quality-Testing)	2027.01.11 - 2027.03.25	1.5 months, 45 days	- Bug fixes, performance, security testing, conference prep.
Phase 3 (Delivery)	2027.04.05	1 - 2 days	- Product Release

Part 2 - Structure

2.1 Required Completion and Steps

Frontend Developer and Project Lead

Deadline: 2027.01.11 (Feature Freeze)

October 2026 (Weeks 1-4)

- Week 1 (Sep 28 – Sep 31): Set up Flutter project structure. Build the shared design system (pink gradients, glassmorphism, typography, spacing). Set up API client with auth token handling. Set up CI/CD.

- Week 2 (Oct 8 - 12): Build login/OTP flow. Build the app shell with empty screens for Teacher and Parent. Connect to the backend /auth/* endpoints.

- Week 3 (Oct 12–18): Build the Quadra Technology bar (static version). The black pill that expands and collapses. No AI yet. Just the UI.

- Week 4 (Oct 19–25): Build the navigation between screens. Teacher Profile, Lesson Planning, Files, Class Analytics, ProLeaf Intelligence (near future)

November 2026 (Weeks 5 - 8)

- Week 5 (Oct 26 – Nov 1): Build the Motion Fluid component. The draggable slider for switching chart types. Connect to mock data.

- Week 6 (Nov 2–8): Build the split-screen workspace. Lesson plan on the left, student data on the right. Build the AI checklist UI (static).

- Week 7 (Nov 9–15): Build the teacher workflow: bulk grade entry, class rosters, student history. Connect to backend /teacher/* endpoints.

- Week 8 (Nov 16–22): Build the parent portal: progress view, comparison charts (their child vs class average vs median academic performance), consent management. Connect to backend /parent/* endpoints.

December 2026 (Weeks 9 - 12)

· Week 9 (Nov 23–29): Connect the Motion Fluid to real data. Build the AI checklist UI with real-time updates (WebSocket/polling). Connect to the AI integration.

· Week 10 (Nov 30 – Dec 6): Polish the Quadra Technology animations. Expand/collapse. Smooth transitions. Fix accessibility issues (text contrast).

· Week 11 & 12 (Dec 7–20): Bug fixes. Performance optimization. Ensure the app runs smoothly on android tablets.

January 2027 (Weeks 13 - 15)

- Week 13 (Dec 21–27): Final polish. Ensure all screens are pixel-perfect. Ensure all animations are smooth.

- Week 14 (Dec 28 – Jan 3): Integration testing with the backend. Fix any API contract issues. Ensure auth works end-to-end.

- Week 15 (Jan 4–11): Feature freeze. Hand over to quality testing. Write test cases. Document known issues.

Quality Testing (Jan 11 – Feb 25)

Fix bugs found during testing:

- Prepare the app for app store submission.

- Final polish for delivery.

- "Volunteer Teachers" for Product Testing.

Backend Developer (Remote, Part-time, Weekends Only)

Deadline: 11 January 2027 (Feature Freeze)

October 2026 (Weekends 1 – 4)

- Weekend 1 (Oct 3–4): API documentation (OpenAPI/Swagger). Version the API (/v1/). Define the error envelope.

- Weekend 2 (Oct 10–11): Build the mock server for the frontend. Realistic fake data for all endpoints.

- Weekend 3 (Oct 17–18): Expand the admin API. School management, class management, student management, account management.

- Weekend 4 (Oct 24–25): Enrollment code lifecycle. Generate, distribute, expire, revoke codes for parents.

November 2026 (Weekends 5 – 8)

- Weekend 5 (Oct 31 – Nov 1): CSV import pipeline. Endpoint, validation, error reporting, transaction safety.

- Weekend 6 (Nov 7–8): CSV import pipeline (continued). Audit logging. Downloadable template.

- Weekend 7 (Nov 14–15): Security hardening. Rate limiting on /auth/otp/request. Token scoping review.

- Weekend 8 (Nov 21–22): Security hardening (continued). Penetration testing. Attempt to break your own authorization boundaries.

December 2026 (Weekends 9 – 12)

- Weekend 9 (Nov 28–29): Database migrations for any new features. Index optimization. Query performance review.

- Weekend 10 (Dec 5–6): Support Developer 2 on AI integration. Review AI data flow. Ensure AI respects authorization boundaries.

- Weekend 11 (Dec 12–13): Core compliance. Data localization (servers in mainland China). Encryption at rest. Secrets management.

- Weekend 12 (Dec 19–20): Monitoring & reliability. Health checks, error tracking (Sentry), alerts for 5xx errors.

January 2027 (Weekends 13 – 15)

- Weekend 13 (Dec 26–27): Backup & recovery. Build the database backup strategy. Test restore.

- Weekend 14 (Jan 2–3): Integration testing with the frontend. Fix any API contract issues.

- Weekend 15 (Jan 9–10): Feature freeze. Hand over to quality testing. Document known issues.

Quality Testing (Jan 11 – Feb 25)

- Fix critical bugs found during testing.

- Support the frontend developer with API issues.

- Ensure the production environment is stable.

Junior Backend Developer

Deadline: 11 January 2027 (Feature Freeze)

October 2026 (Weeks 1-4)

- Week 1 (Sep 28 – Sep 31): Learn the codebase. Read the README. Understand the authorization model, audit system, and database schema. Set up your local environment.

- Week 2 (Oct 8–12): Learn the codebase (continued). Write small features under supervision. Fix bugs. Write tests.

- Week 3 (Oct 12–18): Mainland Cloud Server and ICP Filing Materials and Preparation. Study the intelligence interface. Understand what the AI integration must do. Research LLM providers (OpenAI, Anthropic, DeepSeek, local models).

- Week 4 (Oct 19–25): Implement the intelligence interface with a real LLM provider. Start with a simple prompt. Test it locally.

November 2026 (Weeks 5-8)

- Week 5 (Oct 26 – Nov 1): Prompt engineering. Design prompts for lesson plan generation. Test with real curriculum examples.

- Week 6 (Nov 2–8): Prompt engineering (continued). Design prompts for live classroom checklists. Test with real classroom scenarios.

- Week 7 (Nov 9–15): AI output validation. Build a validation layer so the AI cannot hallucinate dangerous or incorrect content.

- Week 8 (Nov 16–22): Content moderation. Route all AI output through a moderation system before it reaches the teacher.

December 2026 (Weeks 9-12)

- Week 9 (Nov 23–29): MLPS Filing Materials and Preparation. AI cost control. Rate limiting, usage tracking, cost caps for LLM calls.

- Week 10 (Nov 30 – Dec 6): AI fallback logic. Ensure the system degrades gracefully if the AI is slow or unavailable.

- Week 11 (Dec 7–13): Stats pipeline. Expand cmd/statsjob for production. Scheduled execution. More aggregations.

- Week 12 (Dec 14–20): Real-time sync. WebSocket or polling for the AI checklist updates.

January 2027 (Weeks 13 - 15)

- Week 13 (Dec 21–27): Educational APP and Algorithm Filing Materials and Preparation. Compliance infrastructure. AI audit logging. User reporting mechanism for inappropriate AI content.

- Week 14 (Dec 28 – Jan 3): Integration testing with the frontend. Ensure the AI checklist UI works end-to-end.

- Week 15 (Jan 4–11): Feature freeze. Hand over to quality testing. Document known issues.

Quality Testing (Jan 11 - Feb 25)

- Fix bugs found during testing.

- Support the frontend developer with AI endpoint issues.

- Ensure the AI is stable and reliable.

For further questions, please communicate with the General Manager (总经理).